

Phase 7 - Project Documentation

PocketSmart AI: Your Smart Budget & Recommendation Assistant

Date: 29 September 2026

1. Purpose

This phase records the technical structure, setup, modules, routes, user workflow and maintenance guidance for PocketSmart AI.

2. Project Description

PocketSmart AI provides budget-aware recommendations for home interiors, party planning and jewelry selection using FastAPI, Gemini 1.5 Flash Pro, frontend technologies, authentication/session handling and a database layer.

3. Technology Stack

Technology	Use
FastAPI	Backend routing and API endpoints
Gemini 1.5 Flash Pro	Multimodal recommendation processing
HTML/CSS + JavaScript	Interactive frontend
Jinja2	Template rendering
Database	Application persistence
Platform APIs/simulated sourcing	Product and service data

Project Documentation - Structure

4. Root Structure

Path	Purpose
.env	Private environment configuration
.env.example	Example configuration variables
.gitignore	Version-control exclusions
requirements.txt	Python dependencies
README.md	Setup and usage reference
data/pocketSMART.db	Application database
tests/test_app.py	Application tests
app/	Main application package

5. Application Structure

- app/auth.py – authentication.
- app/config.py – configuration.
- app/database.py – database.
- app/main.py – application entry.
- app/models/schemas.py – schemas.
- app/routers/ – authentication, pages and planners.
- app/services/ – catalog, Gemini utilities and recommendations.
- app/static/ – CSS and JavaScript.
- app/templates/ – HTML/Jinja2 pages.

Project Documentation - User Workflow and Routes

6. User Workflow

- Open landing page.
- Register or log in.
- Open dashboard.
- Choose Home, Party or Jewelry Planner.
- Enter budget and contextual information.
- Submit request.
- Review AI recommendations.
- Open History to review previous queries/results.

7. Important Routes

Route	Description
/register	New user registration
/login	Authentication
/logout	Session termination
/token	JWT token issuance
/session-info	Session metadata
/session-data	Session-specific data
/generate-home	Home recommendations
/generate-party	Party recommendations
/generate-jewelry	Jewelry recommendations
/recommendations-details	Detailed recommendations
/history	Past queries/results

Project Documentation - Maintenance

8. Configuration Guidelines

- Keep API keys in environment configuration.
- Do not hard-code private credentials.
- Keep .env private.
- Use .env.example for placeholders.
- Install dependencies from requirements.txt.

9. Maintenance Guidelines

- Keep planner route logic in routers/planners.py.
- Keep AI communication in services/gemini_utils.py.
- Keep recommendation logic in services/recommendation_service.py.
- Keep catalog data in services/catalog.py.
- Keep frontend assets in static/templates.
- Run tests after major changes.

10. Documentation Outcome

This phase provides a reference for architecture, project structure, routes, user workflow, configuration and maintenance.